

Analysis of Medium Access Control: Pure ALOHA and Slotted ALOHA

Summary

This briefing document provides a technical explain of the ALOHA protocols used in Medium Access Control (MAC), specifically focusing on Pure ALOHA and Slotted ALOHA. The analysis highlights the transition from the unrestricted transmission model of Pure ALOHA to the synchronized, time-slotted approach of Slotted ALOHA.

The critical takeaway is that by restricting transmission to the beginning of discrete time slots, Slotted ALOHA reduces the "vulnerable time"—the period during which collisions can occur—by half. This structural improvement results in a doubling of maximum theoretical throughput, increasing it from 18.4% in Pure ALOHA to 36.8% in Slotted ALOHA. The document further details the mathematical foundations for calculating throughput, the impact of frame generation rates (G), and practical numerical applications based on standard network parameters.

1. Fundamentals of ALOHA Protocols

ALOHA is a Medium Access Control protocol where multiple stations share a common channel. The efficiency of these protocols is measured by their **Throughput (S)**, which represents the average number of frames successfully transmitted per frame transmission time.

1.1 Core Definitions

- **Frame Transmission Time (T_f):** The time required to transmit a single frame, calculated as: $T_f = \frac{\text{Frame Size (L)}}{\text{Bandwidth (B)}}$
 - **Vulnerable Time:** The time interval during which if two or more stations transmit, a collision will occur.
 - **G (Load):** The average number of frames generated by the network (all stations combined) during one frame transmission time (T_f).
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2. Pure ALOHA

Pure ALOHA allows stations to transmit data whenever they have a frame ready. While simple to implement, this lack of coordination leads to a high probability of collisions.

2.1 Vulnerable Time in Pure ALOHA

The vulnerable time for Pure ALOHA is $2 \times T_f$. If a station begins transmitting at time t , a collision will occur if:

1. Any other station started transmitting between $t - T_f$ and t (its end would collide with the beginning of the current frame).
2. Any other station starts transmitting between t and $t + T_f$ (its beginning would collide with the end of the current frame).

2.2 Throughput and Efficiency

The throughput for Pure ALOHA is governed by the formula: $S = G \times e^{-2G}$

- **Maximum Throughput:** Occurs when the load $G = 1/2$.
- **Peak Efficiency:** Approximately **18.4%** ($1/2e$).
- **Interpretation:** At maximum efficiency, if 1,000 frames are generated in one T_f , only 184 will be delivered successfully. To achieve this, the network should ideally generate one frame every two frame transmission times ($2T_f$).

3. Slotted ALOHA

Slotted ALOHA was developed to improve the efficiency of Pure ALOHA by introducing time synchronization.

3.1 Mechanism

The shared channel's time is divided into discrete intervals called **slots**, each equal to T_f .

- **Transmission Rule:** A station can only start transmitting at the absolute beginning of a time slot.
- **Delay Requirement:** If a station's data is ready in the middle of a slot, it must wait until the beginning of the next slot to transmit.

3.2 Vulnerable Time in Slotted ALOHA

Because transmissions are restricted to slot boundaries, the vulnerable time is reduced to T_f . Collisions only occur if two or more stations attempt to transmit at the beginning of the exact same slot.

3.3 Throughput and Efficiency

The throughput for Slotted ALOHA is governed by the formula: $S = G \times e^{-G}$

- **Maximum Throughput:** Occurs when the load $G = 1$.
- **Peak Efficiency:** Approximately **36.8%** ($1/e$).
- **Interpretation:** Reducing the vulnerable time by half doubles the maximum throughput compared to Pure ALOHA. At peak efficiency, if 1,000 frames are generated, 368 will be delivered successfully. This is achieved when the network generates exactly one frame per frame transmission time (T_f).

4. Comparative Summary

The following table summarizes the key technical differences between the two protocols:

Feature	Pure ALOHA	Slotted ALOHA
Transmission Time	Any time	Beginning of a slot only
Vulnerable Time	$2 \times T_f$	T_f
Throughput Formula	$S = G \times e^{-2G}$	$S = G \times e^{-G}$
Max Throughput ($S_{\max}$)	18.4%	36.8%
G for $S_{\max}$	0.5	1.0
Implementation	Simple (no synchronization)	Complex (requires synchronization)

5. Numerical Application Analysis

Using a standardized scenario (Frame Size = 200 bits, Bandwidth = 200 kbps), the frame transmission time (T_f) is calculated as **1 millisecond (ms)**.

5.1 Throughput at Variable Loads

The performance of the protocols varies significantly based on the number of frames generated per second across the network:

Network Load (Frames/sec)	G (Frames per T_f)	Pure ALOHA Successes	Slotted ALOHA Successes
1,000	$G = 1.0$	~135 frames/sec	~368 frames/sec
500	$G = 0.5$	~92 frames/sec	~303 frames/sec
250	$G = 0.25$	~38 frames/sec	~195 frames/sec

5.2 Key Observations from Data

- Load Impact:** In Pure ALOHA, $G=1.0$ results in a lower success rate (135) than $G=0.5$ (92 is the count, but the *efficiency* is higher at 0.5), because $G=1.0$ exceeds the optimal load for that protocol, leading to excessive collisions.
- Peak Performance:** For Slotted ALOHA, $G=1.0$ is the optimal operating point, yielding the maximum 368 successful frames per second.
- Collision Waste:** In all scenarios, a significant portion of generated frames are lost to collisions. For example, in Pure ALOHA at $G=1.0$, 865 out of 1,000 frames are lost.